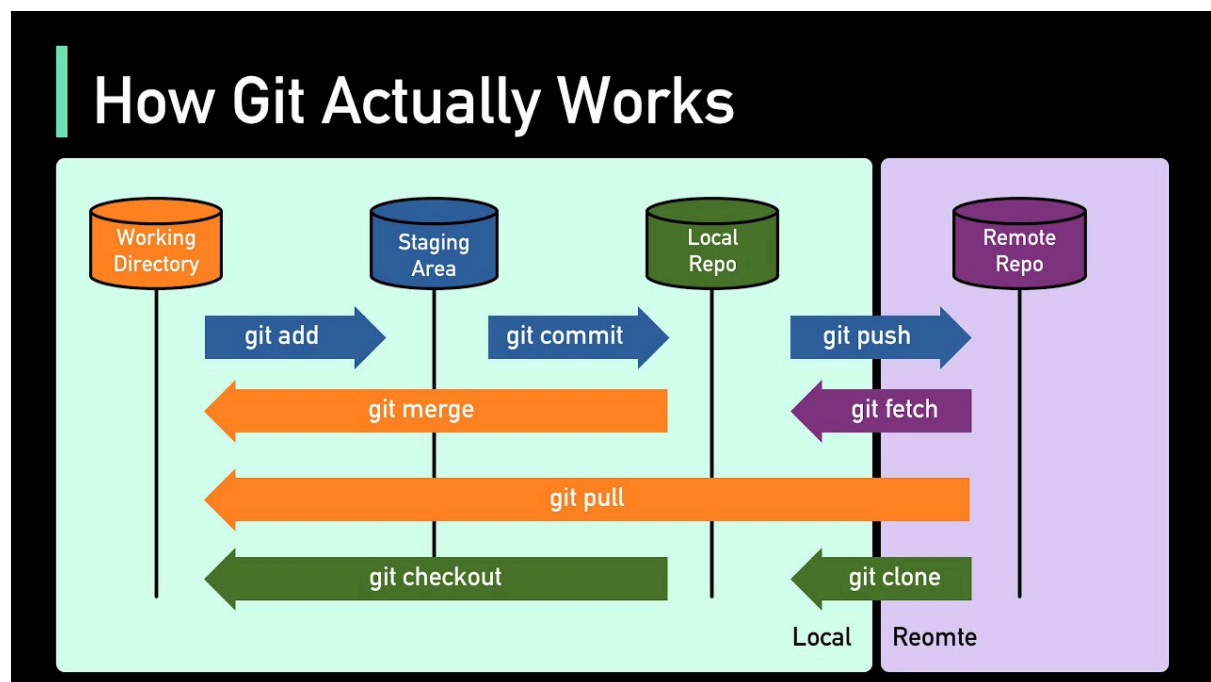


GIT

Git is an open-source distributed version control system. It is designed to handle minor to major projects with high speed and efficiency. It is developed to coordinate the work among the developers. The version control allows us to track and work together with our team members at the same workspace. Git is the foundation of many services like GitHub and GitLab, but we can use Git without using any other Git services. Git can be used privately and publicly. Most popular version control management and scalable repository management system which is open source, fast, and lightweight and easy to use compared to other version control management systems like mercurial etc.

Features of Git

Some remarkable features of Git are as follows:



- Open Source, Scalable, Distributed
- Security, Speed
- Supports non-linear development
- Branching and Merging, Data Assurance

There are 3 stages in git repository

1. Modified (still in local repository) [LOCAL REPO]
2. Staged (ready for commit) [LOCAL REPO]
3. Committed (which gets pushed into online repository) [LOCAL REPO]

4. Pushed Stage(push command loads the changes to the online repo)
[REMOTE REPO]

GIT COMMANDS

- **git clone <url>**
 - Used to create clone of the online repository to you local machine
- **git status**
 - Status of local repo and how repo looks in comparison to the online repository
- **git -v**
 - Used to check the current git version
- **git branch**
 - To check branch that is currently selected
- **git add .**
 - To move all the files from local machine to the staging area
- **git add <filename>**
 - To move the specified file from local machine to the staging area
- **git reset**
 - To remove all the files that have been added to the staging area
- **git reset <filename>**
 - To remove a file that has been added to the staging area
- **git stash**
 - To remove changes that have not been staged from local machine
- **git commit -m "commit message"**
 - To move all the staged files from staging area to local repository
- **git push**
 - To move the files in local repository to online repository
- **git push --set-upstream origin <branch name>**
 - To create a new branch and set upstream as the specified branch
- **git pull**
 - To download and merge all files from remote repository to the local machine
- **git fetch**
 - To download and merge files from remote repository to local repository
- **git pull origin <branchname>**
 - To create a pull request to merge from specified branch to the currently located branch
- **git checkout <branchname>**
 - To move to a specified branch that is already present
- **git checkout -b <branchname>**
 - To create and move to a specified new branch